

Subtraction of integers

Subtracting is adding the opposite — one rule that replaces the whole operation.

LESSON 27–28

2 × 40 MIN

MATEMATIKA 7, §11

S8 1.2

LESSON CLOCK

15'

25'

25'

20'

5'

One rule	15 min
Two signs in a row	25 min
Chains	25 min
Differences	20 min
Homework	5 min

LEARNING OBJECTIVES

- State and use $a - b = a + (-b)$.
- Simplify expressions containing two signs in a row.
- Compute chains of additions and subtractions.
- Use subtraction to find a difference or a change.

TEXTBOOK

National	pp. 71–76
Cambridge	Stage 8, pp. 6–10
Workbook	Exercise 1.2

01

Explanation

One rule

$$a - b = a + (-b)$$

To subtract a number, add its opposite. Subtraction is not a new operation at all — it is addition, once the opposite is taken.

SUBTRACTION	AS AN ADDITION	ANSWER
$7 - 3$	$7 + (-3)$	4
$3 - 7$	$3 + (-7)$	-4
$-3 - 7$	$-3 + (-7)$	-10
$-3 - (-7)$	$-3 + 7$	4
$3 - (-7)$	$3 + 7$	10

NOTHING NEW HAS TO BE LEARNT

Every subtraction becomes an addition, and addition was the last lesson. That is why the rule for subtraction is a single line while the rule for addition took two cases.

Two signs in a row

WRITTEN	MEANS	EXAMPLE
$+(+a)$	$+a$	$5 + (+3) = 8$
$+(-a)$	$-a$	$5 + (-3) = 2$
$-(+a)$	$-a$	$5 - (+3) = 2$
$-(-a)$	$+a$	$5 - (-3) = 8$

Two like signs give a plus; two unlike signs give a minus.

$5 - (-3)$ IS 8, NOT 2

Taking away a debt leaves you better off. The everyday sentence explains the arithmetic, and it is worth saying aloud each time until the rule is automatic.

Chains

Rewrite every subtraction as an addition first, then use the two-pile method.

Example. $-4 - (-9) + (-6) - 2$.

STEP	EXPRESSION
rewrite	$-4 + 9 - 6 - 2$
positives	9
negatives	$-4 - 6 - 2 = -12$
combine	$9 - 12 = -3$

REWRITE THE WHOLE CHAIN BEFORE COMPUTING ANYTHING

Doing the sign work and the arithmetic at the same time is where errors come from. Two separate passes are quicker in the end.

Differences

The **change** from a to b is $b - a$: the new value minus the old one. A positive answer is a rise, a negative one a fall.

FROM	TO	CHANGE	MEANING
4°	11°	$+7$	a rise of 7
4°	-3°	-7	a fall of 7
-9°	-2°	$+7$	a rise of 7
-2°	-9°	-7	a fall of 7

“NEW MINUS OLD”, IN THAT ORDER

Reversing the subtraction reverses the sign and turns a rise into a fall. Writing the words “new – old” at the top of the working prevents it.

02

Worked examples

EXAMPLE 1 Compute $3 - (-7)$, $-3 - 7$ and $-3 - (-7)$.

- 1

$3 - (-7) = 3 + 7 = 10$
Two minuses give a plus.
- 2

$-3 - 7 = -3 + (-7) = -10$
- 3

$-3 - (-7) = -3 + 7 = 4$
- 4

Rewrite first, compute second.

ANSWER
 $10, -10, 4$

EXAMPLE 2 Compute $-4 - (-9) + (-6) - 2$.① Rewrite: $-4 + 9 - 6 - 2$.

② Positives: 9.

③ Negatives: -12 .④ $9 - 12 = -3$

ANSWER

 -3 **EXAMPLE 3** The temperature falls from 4°C to -9°C . Find the change and the size of the fall.① Change = $\text{new} - \text{old} = -9 - 4$.② $= -13$

③ Negative, so it is a fall.

④ A fall of 13 degrees.

ANSWER

 -13 — a fall of 13° **03 Interactive model**

Say “taking away a debt makes you richer” once, and then let the class supply it themselves every time a $-(-)$ appears; the rule sticks as a sentence, not a symbol.

Subtracting on the line

$$x > 2$$

inequality $x > 2$ boundary **2 (open)** $x = 0$ works? **no** $x = 6$ works? **yes**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Subtraction	Ayirish	Вычитание
Difference	Ayirma	Разность
Minuend	Kamayuvchi	Уменьшаемое
Subtrahend	Ayiriluvchi	Вычитаемое
Opposite	Qarama-qarshi	Противоположное
Two signs in a row	Ketma-ket ikki ishora	Два знака подряд
Change	O'zgarish	Изменение
Rise and fall	Ko'tarilish va pasayish	Подъём и спад

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$a - b$ equals:

$5 - (-3)$ equals:

$-3 - 7$ equals:

$-(-a)$ equals:

$-a$

a

0

$|a|$

The change from 4 to -3 :

7

-7

1

-1

A negative change means:

a rise

a fall

no change

an error

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $7 - 3$

ANSWER 4

2. $3 - 7$

ANSWER -4

3. $-3 - 7$

ANSWER -10

4. $3 - (-7)$

ANSWER 10

5. $-3 - (-7)$

ANSWER 4

6. $5 - (+3)$

ANSWER 2

7. $5 + (-3)$

ANSWER 2

Medium · the standard question

7 PROBLEMS

1. $-4 - (-9) + (-6) - 2$

ANSWER -3

2. $12 - 20 + 5 - (-3)$

ANSWER 0

3. $-15 - (-15)$

ANSWER 0

4. $0 - (-8)$

ANSWER 8

5. Change from 4° to -9°

ANSWER -13

6. Change from -9° to -2°

ANSWER $+7$

7. $-7 - (-2) - 5$

ANSWER -10

Hard · stretch and reason

7 PROBLEMS

1. $-3 - (-5) - (-7) - 9$

ANSWER 0

2. Find x : $x - (-4) = 1$

ANSWER -3

3. Find x : $-6 - x = 2$

ANSWER -8

4. The difference between the highest and lowest of $-12, 5, -3, 8$

ANSWER 20

5. A lift from -3 to 9 : floors travelled

ANSWER 12

6. On a day the temperature goes $-7 \rightarrow 3 \rightarrow -5$

ANSWER A rise of 10, a fall of 8

7. Simplify $a - (-b) - (-c)$

ANSWER $a + b + c$

07 Homework

Homework — 5 tasks

Rewrite every subtraction as an addition before computing anything.

1. Compute $8 - (-5)$, $-8 - 5$ and $-8 - (-5)$.
2. Compute $-6 - (-11) + (-4) - 3$.
3. The temperature falls from 7°C to -6°C . Find the change.
4. Find the difference between the highest and lowest of $-15, 4, -8, 11$.
5. Find x if $x - (-6) = -1$.